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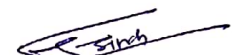
UNIT: III

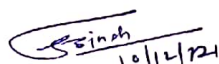
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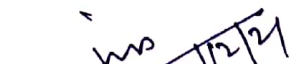
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References:

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons


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ELEMENT OF MECHANICAL ENGINEERING

LECTURE - 17 (UNIT-03)

Syllabus:- INTRODUCTION OF GAS POWER CYCLES, AIR STANDARD EFFICIENCY

GAS POWER CYCLES

Here gas is the working fluid. It does not undergo any phase change. Engines operating on gas cycles may be either cyclic or non-cyclic. Hot air engines using air as the working fluid operate on a closed cycle. Internal combustion engines where the combustion of fuel takes place inside the engine cylinder are noncyclic heat engines.

The various Gas Power Cycles are as follows:

1. Carnot Cycle
2. Stirling Cycle
3. Ericsson Cycle

CARNOT CYCLE

The Carnot cycle consists of two reversible isotherms and two reversible adiabatics. If an ideal gas is assumed as the working fluid. Then for 1kg of gas,

$$\begin{aligned}
 Q_{1-2} &= RT_1 \ln \frac{v_2}{v_1}; & W_{1-2} &= RT_1 \ln \frac{v_2}{v_1} \\
 Q_{2-3} &= 0; & W_{2-3} &= -c_v(T_3 - T_2) \\
 Q_{3-4} &= RT_2 \ln \frac{v_4}{v_3}; & W_{3-4} &= RT_2 \ln \frac{v_4}{v_3} \\
 Q_{4-1} &= 0; & W_{4-1} &= -c_v(T_1 - T_4)
 \end{aligned}$$

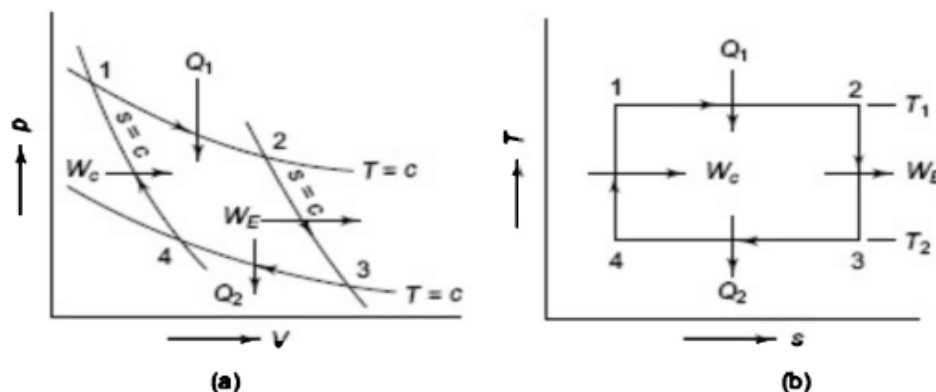


Fig.: Carnot Cycle

$$\therefore \sum_{\text{cycle}} dQ = \sum_{\text{cycle}} dW$$

Now,

$$\frac{v_2}{v_3} = \left(\frac{T_2}{T_1}\right)^{1/(\gamma-1)}$$

and

$$\frac{v_1}{v_4} = \left(\frac{T_2}{T_1}\right)^{1/(\gamma-1)}$$

$$\frac{v_2}{v_3} = \frac{v_1}{v_4} \text{ or } \frac{v_2}{v_1} = \frac{v_3}{v_4}$$

Therefore,

$$Q_1 = \text{Heat added} = RT_1 \ln \frac{v_2}{v_1}$$

$$W_{\text{net}} = Q_1 - Q_2 = R \ln \frac{v_2}{v_1} \cdot (T_1 - T_2)$$

$$\therefore \eta_{\text{cycle}} = \frac{W_{\text{net}}}{Q_1} = \frac{T_1 - T_2}{T_1}$$

The large back work ($W_c = W_{4-1}$) is a big drawback for the Carnot gas cycle, as in the case of the Carnot vapour cycle.

STIRLING CYCLE

The Stirling cycle consists of two reversible isotherms and two reversible isochores. For 1 kg of ideal gas

$$Q_{1-2} = W_{1-2} = RT_1 \ln \frac{v_2}{v_1}$$

$$Q_{2-3} = c_v(T_2 - T_1); W_{2-3} = 0$$

$$Q_{3-4} = W_{3-4} = -RT_2 \ln \frac{v_3}{v_4}$$

$$Q_{4-1} = c_v(T_1 - T_2); W_{4-1} = 0$$

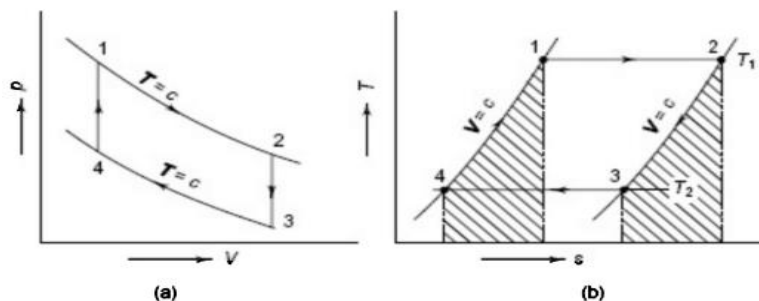


Fig.: Stirling Cycle

Due to heat transfers at constant volume processes, the efficiency of the Stirling cycle is less than that of the Carnot cycle. However, if a regenerative arrangement is used such that

$Q_{2-3} = Q_{4-1}$, i.e., the area under 2-3 is equal to the area under 4-1, then the cycle efficiency becomes

$$\eta = \frac{RT_1 \ln \frac{v_2}{v_1} - RT_2 \ln \frac{v_3}{v_4}}{RT_1 \ln \frac{v_2}{v_1}} = \frac{T_1 - T_2}{T_1}$$

So, the regenerative Stirling cycle has the same efficiency as the Carnot cycle.

ERICSSON CYCLE

The Ericsson cycle is made up of two reversible isotherms and two reversible isobars.

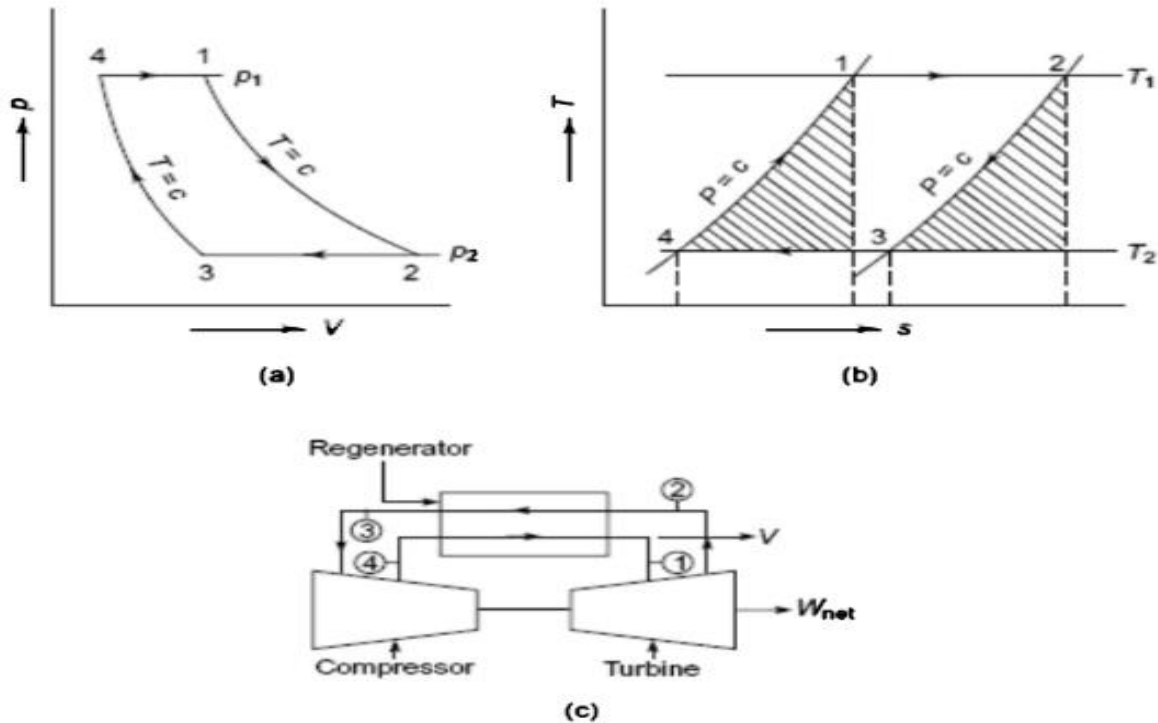


Fig.: Ericsson Cycle

For 1 kg of ideal gas

$$\begin{aligned} Q_{1-2} &= W_{1-2} = RT_1 \ln \frac{p_1}{p_2} \\ Q_{2-3} &= c_p(T_2 - T_1); W_{2-3} = p_2(v_3 - v_2) = R(T_2 - T_1) \\ Q_{3-4} &= W_{3-4} = -RT_2 \ln \frac{p_1}{p_2} \\ Q_{4-1} &= c_p(T_1 - T_4); W_{4-1} = p_1(v_1 - v_4) = R(T_1 - T_2) \end{aligned}$$

Since part of the heat is transferred at constant pressure and part at constant temperature, the efficiency of the Ericsson cycle is less than that of the Carnot cycle. But with ideal regeneration, $Q_{2-3} = Q_{4-1}$ so that all the heat is added from the external source at T_1 and all the heat is rejected to an external sink at T_2 , the efficiency of the cycle becomes equal to the Carnot cycle efficiency, since

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{RT_2 \ln \frac{p_1}{p_2}}{RT_1 \ln \frac{p_1}{p_2}} = 1 - \frac{T_2}{T_1}$$

The regenerative, Stirling and Ericsson cycles have the same efficiency as the Carnot cycle, but much less back work. Both cycles utilise regeneration, a process during which heat is transferred to a thermal energy storage device called a regenerator during one part of the cycle and is transferred back to the working fluid during another part of the cycle. Hot air engines working on these cycles have been successfully operated. But it is difficult to transfer heat to a gas at high rates, since the gas film has a very low thermal conductivity. So there has not been much progress in the development of hot air engines. However, since the cost of internal combustion engine fuels is getting excessive, these may find a field of use in the near future.

REFERENCES:-

1. P.K. Nag, Engineering Thermodynamics, Mc Graw Hill
2. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
3. B. Agrawal, Mechanical Engineering, John Wiley & Sons

ELEMENT OF MECHANICAL ENGINEERING

LECTURE - 18 (UNIT-03)

Syllabus:- AIR STANDARD EFFICIENCY OF OTTO & DIESEL CYCLE

AIR STANDARD EFFICIENCY

The efficiency of the engine using air as the working medium is known as an "Air standard efficiency". This efficiency is often called ideal efficiency.

AIR STANDARD EFFICIENCY OF OTTO & DIESEL CYCLE

Working cycle:

(a) Otto cycle- thermodynamic cycle for SI/petrol engine

-Reversible adiabatic compression and expansion process

-Constant volume heat addition (combustion) and heat rejection process (exhaust)

Figure 7 depicts the Otto cycle

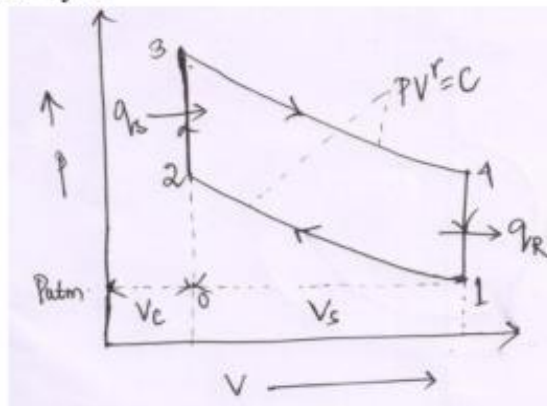


Fig. 7. Otto cycle

Heat supplied, $q_s = C_v(T_3 - T_2)$

Heat rejection, $q_R = C_v(T_4 - T_1)$

Compression ratio, $r_k = \frac{V_1}{V_2}$

Thermal efficiency, $\eta_{th} = \frac{q_s - q_R}{q_s} = \frac{C_v(T_3 - T_2) - C_v(T_4 - T_1)}{C_v(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$

In process 1-2, adiabatic compression process,

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$\Rightarrow T_2 = T_1 \cdot (r_k)^{\gamma-1}$$

In adiabatic expansion process, i.e. 3-4,

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\Rightarrow T_3 = T_4 \cdot (r_k)^{\gamma-1}$$

$$\eta_{th} = 1 - \frac{T_4 - T_1}{T_4 \cdot (r_k)^{\gamma-1} - T_1 \cdot (r_k)^{\gamma-1}}$$

$$= 1 - \frac{1}{(r_k)^{\gamma-1}}$$

Work done (W)

Pressure ratio, $r_p = \frac{P_3}{P_2} = \frac{P_4}{P_1}$

$$\frac{P_2}{P_1} = \frac{P_3}{P_4} = \left(\frac{V_1}{V_2}\right)^\gamma = (r_k)^\gamma$$

$$W = \frac{P_3 V_3 - P_4 V_4}{\gamma - 1} - \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

$$= \frac{1}{\gamma - 1} \left[P_4 V_4 \left(\frac{P_3 V_3}{P_4 V_4} - 1 \right) - P_1 V_1 \left(\frac{P_2 V_2}{P_1 V_1} - 1 \right) \right]$$

$$= \frac{1}{\gamma - 1} [P_4 V_1 (r_k^{\gamma-1} - 1) - P_1 V_1 (r_k^{\gamma-1} - 1)]$$

$$= \frac{P_1 V_1}{\gamma - 1} [r_p (r_k^{\gamma-1} - 1) - (r_k^{\gamma-1} - 1)]$$

$$= \frac{P_1 V_1}{\gamma - 1} [(r_k^{\gamma-1} - 1)(r_p - 1)]$$

Mean effective pressure, $P_m = \frac{\text{work done}}{\text{Swept volume}} = \frac{\text{work done}}{V_1 - V_2}$

$$P_m = \frac{\frac{P_1 V_1}{\gamma - 1} [(r_k^{\gamma-1} - 1)(r_p - 1)]}{V_1 - V_2} = \frac{P_1 r_k [(r_k^{\gamma-1} - 1)(r_p - 1)]}{(\gamma - 1)(r_k - 1)}$$

(b) **Diesel cycle**- thermodynamic cycle for low speed CI/diesel engine

- Reversible adiabatic compression and expansion process
- Constant pressure heat addition (combustion) and heat rejection process (exhaust)

Figure 8 depicts the diesel cycle.

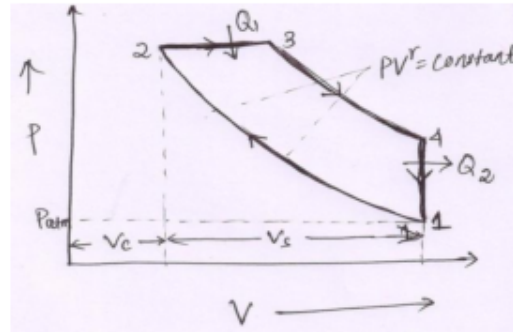


Fig. 8. Diesel cycle

Heat supplied, $Q_1 = C_p(T_3 - T_2)$

Heat rejection, $Q_2 = C_v(T_4 - T_1)$

Compression ratio, $r_k = \frac{V_1}{V_2}$

Cut off ratio, $r_c = \frac{V_3}{V_2}$

Thermal efficiency, $\eta_{th} = \frac{Q_1 - Q_2}{Q_1} = \frac{C_p(T_3 - T_2) - C_v(T_4 - T_1)}{C_p(T_3 - T_2)} = 1 - \frac{1}{\gamma} \frac{(T_4 - T_1)}{(T_3 - T_2)}$

In adiabatic compression process i.e. 1-2,

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$\Rightarrow T_2 = T_1 \cdot (r_k)^{\gamma-1}$$

In process 2-3, pressure constant, then

$$\frac{T_3}{T_2} = \frac{V_3}{V_2} = r_c$$

$$\Rightarrow T_3 = T_2 \cdot r_c = T_1 \cdot (r_k)^{\gamma-1} \cdot r_c$$

In adiabatic expansion process i.e. 3-4,

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} = \left(\frac{V_3}{V_2} \cdot \frac{V_2}{V_4}\right)^{\gamma-1} = (r_c)^{\gamma-1} \cdot \frac{1}{(r_k)^{\gamma-1}}$$

$$\Rightarrow T_4 = T_3 \cdot (r_c)^{\gamma-1} \cdot \frac{1}{(r_k)^{\gamma-1}} = T_1 \cdot (r_k)^{\gamma-1} \cdot r_c \cdot (r_c)^{\gamma-1} \cdot \frac{1}{(r_k)^{\gamma-1}} = T_1 \cdot r_c$$

$$\eta_{th} = 1 - \frac{1}{\gamma} \frac{(T_4 - T_1)}{(T_3 - T_2)} = 1 - \frac{1}{\gamma \cdot (r_k)^{\gamma-1}} \left[\frac{(r_c)^{\gamma} - 1}{r_c - 1} \right]$$

Work done (W)

$$W = P_2(V_3 - V_2) + \frac{P_3 V_3 - P_4 V_4}{\gamma - 1} - \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

$$= P_2(r_c V_2 - V_2) + \frac{P_2 r_c V_2 - P_4 r_k V_2}{\gamma - 1} - \frac{P_2 V_2 - P_1 r_k V_2}{\gamma - 1} \quad \text{since } V_4 = V_1$$

$$= P_2 V_2 \left[\frac{(r_c - 1)(\gamma - 1) + (r_c - r_c^{\gamma} \cdot r_k^{-\gamma} \cdot r_k) - (1 - r_k^{1-\gamma})}{\gamma - 1} \right]$$

$$= P_1 V_1 \cdot r_k^{\gamma-1} \left[\frac{\gamma(r_c - 1) - r_k^{1-\gamma}(r_c^{\gamma} - 1)}{\gamma - 1} \right]$$

REFERENCES:-

1. P.K. Nag, Engineering Thermodynamics, Mc Graw Hill
2. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
3. B. Agrawal, Mechanical Engineering, John Wiley & Sons



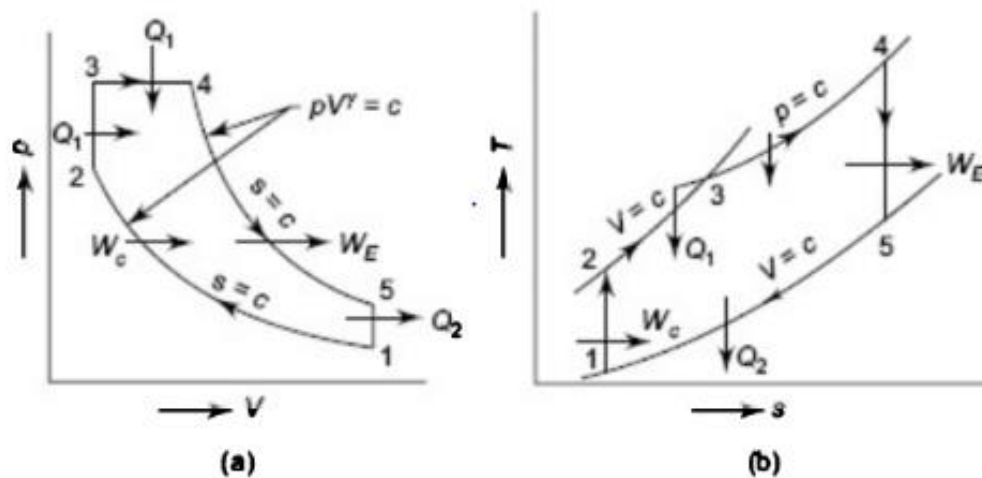
ELEMENT OF MECHANICAL ENGINEERING

LECTURE - 19 (UNIT-03)

Syllabus:- AIR STANDARD EFFICIENCY OF DUAL & BRAYTON CYCLE

MIXED OR DUAL CYCLE

The air standard Diesel cycle does not simulate exactly the pressure-volume variation in an actual compression ignition engine, where the fuel injection is started before the end of compression stroke. A closer approximation is the limited pressure cycle in which some part of heat is added to air at constant volume, and the remainder at constant pressure.



Above figure shows the p - v and T-s diagrams of the dual cycle. Heat is added reversibly, partly at constant volume (2-3) and partly at constant pressure (3-4).

$$\text{Heat supplied } Q_1 = mc_v(T_3 - T_2) + mc_p(T_4 - T_3)$$

$$\text{Heat rejected } Q_2 = mc_v(T_5 - T_1)$$

Efficiency

$$\begin{aligned} \eta &= 1 - \frac{Q_2}{Q_1} \\ &= 1 - \frac{mc_v(T_5 - T_1)}{mc_v(T_3 - T_2) + mc_p(T_4 - T_3)} \\ &= 1 - \frac{T_5 - T_1}{(T_3 - T_2) + \gamma(T_4 - T_3)} \end{aligned}$$

The efficiency of the cycle can be expressed in terms of the following ratios:

Compression Ratio,
$$r_k = \frac{V_1}{V_2}$$

Expansion ratio, $r_e = \frac{V_5}{V_4}$

Cut-off ratio, $r_c = \frac{V_4}{V_3}$

Constant volume pressure ratio, $r_p = \frac{p_3}{p_2}$

It is seen as before that

$$r_k = r_c \cdot r_e$$

or $r_e = \frac{r_k}{r_c}$

Process 3 – 4

$$\frac{p_2 V_2}{T_2} = \frac{p_3 V_3}{T_3}$$

$$T_2 = T_3 \frac{p_2}{p_3} = \frac{T_4}{r_p \cdot r_c}$$

Process 2 - 3

$$\frac{p_2 V_2}{T_2} = \frac{p_3 V_3}{T_3}$$

$$T_2 = T_3 \frac{p_2}{p_3} = \frac{T_4}{r_p \cdot r_c}$$

Process 1-2

$$\frac{T_1}{T_2} = \left(\frac{v_2}{v_1} \right)^{\gamma-1} = \frac{1}{r_k^{\gamma-1}}$$

$$\therefore T_1 = \frac{T_4}{r_p \cdot r_c \cdot r_k^{\gamma-1}}$$

Process 4 – 5

$$\frac{T_5}{T_4} = \left(\frac{v_4}{v_5} \right)^{\gamma-1} = \frac{1}{r_c^{\gamma-1}}$$

$$T_5 = T_4 \cdot \frac{r_c^{\gamma-1}}{r_k^{\gamma-1}}$$

Substituting the values of T_1 , T_2 , T_3 and T_5 in expression of efficiency

$$\eta = 1 - \frac{T_4 \cdot \frac{r_c^{\gamma-1}}{r_k^{\gamma-1}} - \frac{T_4}{r_p \cdot r_c \cdot r_k^{\gamma-1}}}{\left(\frac{T_4}{r_c} - \frac{T_4}{r_p \cdot r_c}\right) + \gamma \left(T_4 - \frac{T_4}{r_c}\right)}$$
$$\eta_{\text{Dual}} = 1 - \frac{1}{r_k^{\gamma-1}} \frac{r_p \cdot r_c^{\gamma} - 1}{r_p - 1 + \gamma r_p (r_c - 1)}$$

REFERENCES:-

1. P.K. Nag, Engineering Thermodynamics, Mc Graw Hill
2. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
3. B. Agrawal, Mechanical Engineering, John Wiley & Sons



ELEMENT OF MECHANICAL ENGINEERING**LECTURE - 20 (UNIT-03)****Syllabus:- AIR STANDARD EFFICIENCY OF BRAYTON OR JOULE CYCLE****BRAYTON OR JOULE CYCLE**

A simple gas turbine power plant is shown in following Fig. Air is first compressed adiabatically in process a-b, it then enters the combustion chamber where fuel is injected and burned essentially at constant pressure in process b-c, and then the products of combustion expand in the turbine to the ambient pressure in process c-d and are thrown out to the surroundings. The cycle is open. The state diagram on the p-v coordinates is shown in Fig. Open cycles are used in aircraft, automotive (buses and trucks) and industrial gas turbine installations.

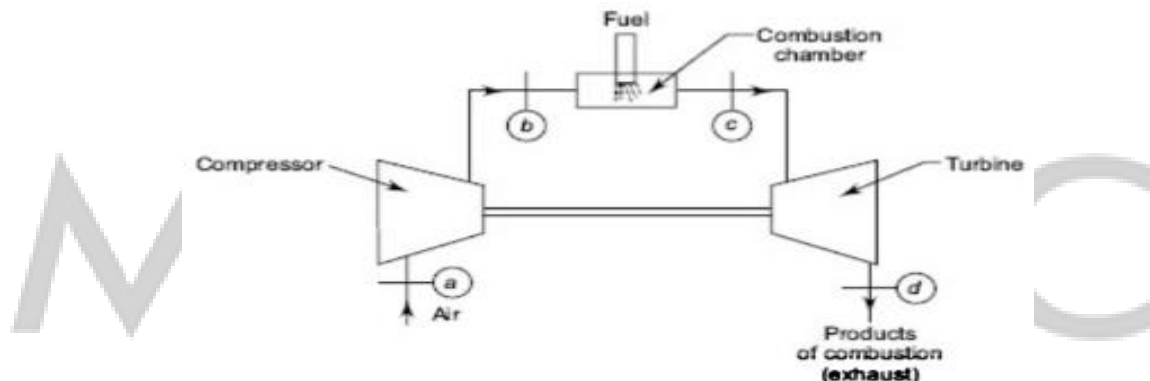


Fig. A Simple Gas Turbine Plant

The Brayton cycle is the air standard cycle for the gas turbine power plant. Here air is first compressed reversibly and adiabatically, heat is added to it reversibly at constant pressure, air expands in the turbine reversibly and adiabatically, and heat is then rejected from the air reversibly at constant pressure to bring it to the initial state. The Brayton cycle, therefore, consists of two reversible isobars and two reversible adiabatics.

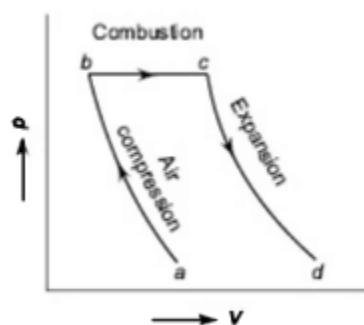


Fig. State diagram of a gas turbine plant in p-v plot

The flow, p-v and T-s diagrams are shown in following Fig. For m kg of air

$$Q_1 = \text{heat supplied} = mc_p(T_3 - T_2)$$

$$Q_2 = \text{heat rejected} = mc_p(T_4 - T_1)$$

$$\begin{aligned} \text{cycle efficiency, } \eta &= 1 - \frac{Q_2}{Q_1} \\ &= 1 - \frac{T_4 - T_1}{T_3 - T_2} \end{aligned}$$

Now,

$$\begin{aligned} \frac{T_2}{T_1} &= \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T_3}{T_4} \quad (\text{Since } p_2 = p_3 \text{ and } p_4 = p_1) \\ \frac{T_4}{T_1} - 1 &= \frac{T_3}{T_2} - 1 \end{aligned}$$

or

$$\frac{T_4 - T_1}{T_3 - T_2} = \frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{(\gamma-1)/\gamma} = \left(\frac{v_2}{v_1}\right)^{\gamma-1}$$

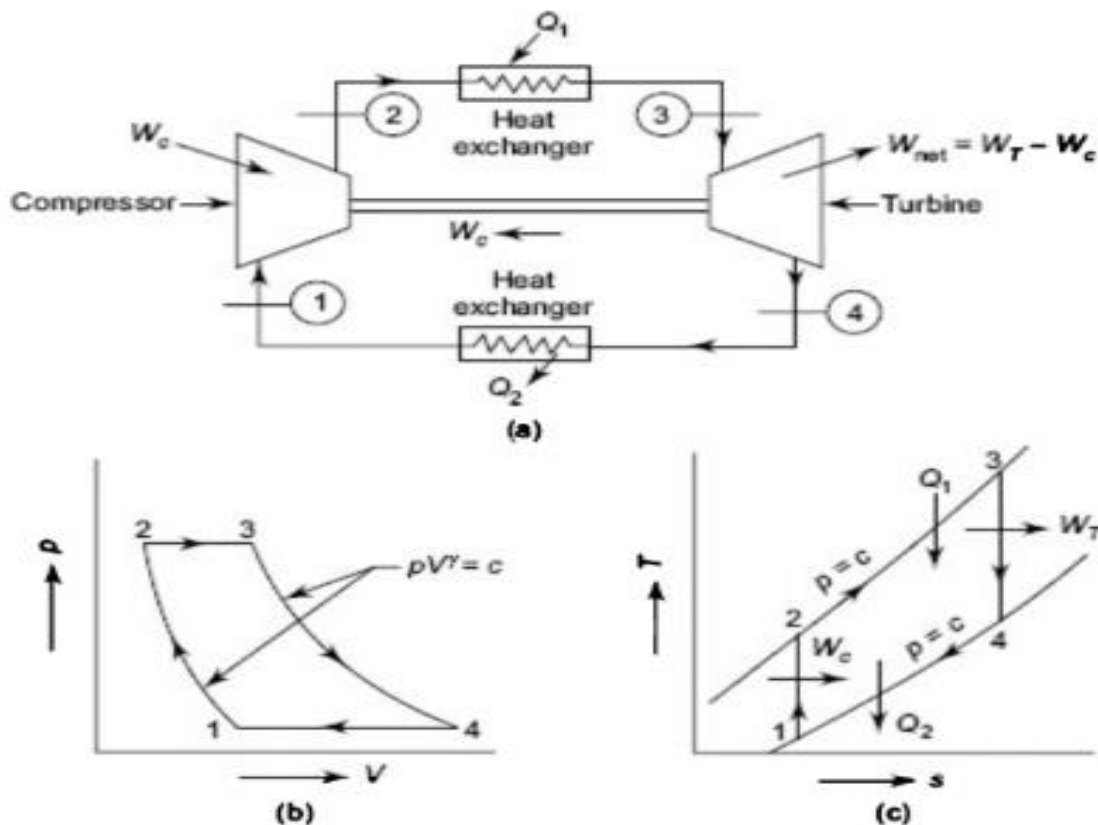


Fig. Brayton Cycle or Joule Cycle

If r_k = compression ratio = v_1/v_2 the efficiency becomes

$$\eta = 1 - \left(\frac{v_2}{v_1}\right)^{\gamma-1}$$

or

$$\eta_{\text{Brayton}} = 1 - \frac{1}{r_k^{\gamma-1}}$$

If r_p = pressure ratio = p_2/p_1 the efficiency may be expressed in the following form also

$$\eta = 1 - \left(\frac{p_1}{p_2}\right)^{(\gamma-1)/\gamma}$$

or

$$\eta_{\text{Brayton}} = 1 - \frac{1}{(r_p)^{(\gamma-1)/\gamma}}$$

The efficiency of the Brayton cycle, therefore, depends upon either the compression ratio or the pressure ratio. For the same compression ratio, the Brayton cycle efficiency is equal to the Otto cycle efficiency.

A closed cycle gas turbine plant is used in a gas-cooled nuclear reactor plant, where the source is a high temperature gas-cooled reactor (HTGR) supplying heat from nuclear fission directly to the working fluid (a gas).

REFERENCES:-

1. P.K. Nag, Engineering Thermodynamics, Mc Graw Hill
2. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
3. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -21 (UNIT-3)**1. INTRODUCTION**

An engine is a device which transforms one form of energy into another form. While transforming energy from one form to another, the efficiency of conversion plays an important role.

Heat engine is a device which transforms the chemical energy of a fuel into the thermal energy and utilizes this thermal energy to perform useful work. Thus, in heat engine, thermal energy is converted into mechanical work.

Heat engines can be broadly classified into two categories:

- (i) External Combustion Engines (EC Engines)
- (ii) Internal Combustion Engines (IC Engines)

(i) External Combustion Engines

In external combustion engine, combustion of fuel takes place outside of the cylinder. For example, in a steam engine or a steam turbine, the heat generated due to the combustion of fuel is employed to generate high pressure steam which is used as the working fluid in a reciprocating engine or a turbine. Other examples are hot air engines, closed cycle gas turbine. These engines are used for driving locomotives, ships, generation of electric power etc.

(ii) Internal Combustion Engines

In internal combustion engine, combustion of fuel with oxygen occurs within the cylinder of the engine. The IC engines group include engines employing mixture of combustible gases and air is known as gas engines. Those using lighter liquid fuels or spirit known as petrol engines and those using heavier liquid fuels, known as oil compression ignition or diesel engines.

2. ENGINE COMPONENT: -

The major component of engine and their function are as following

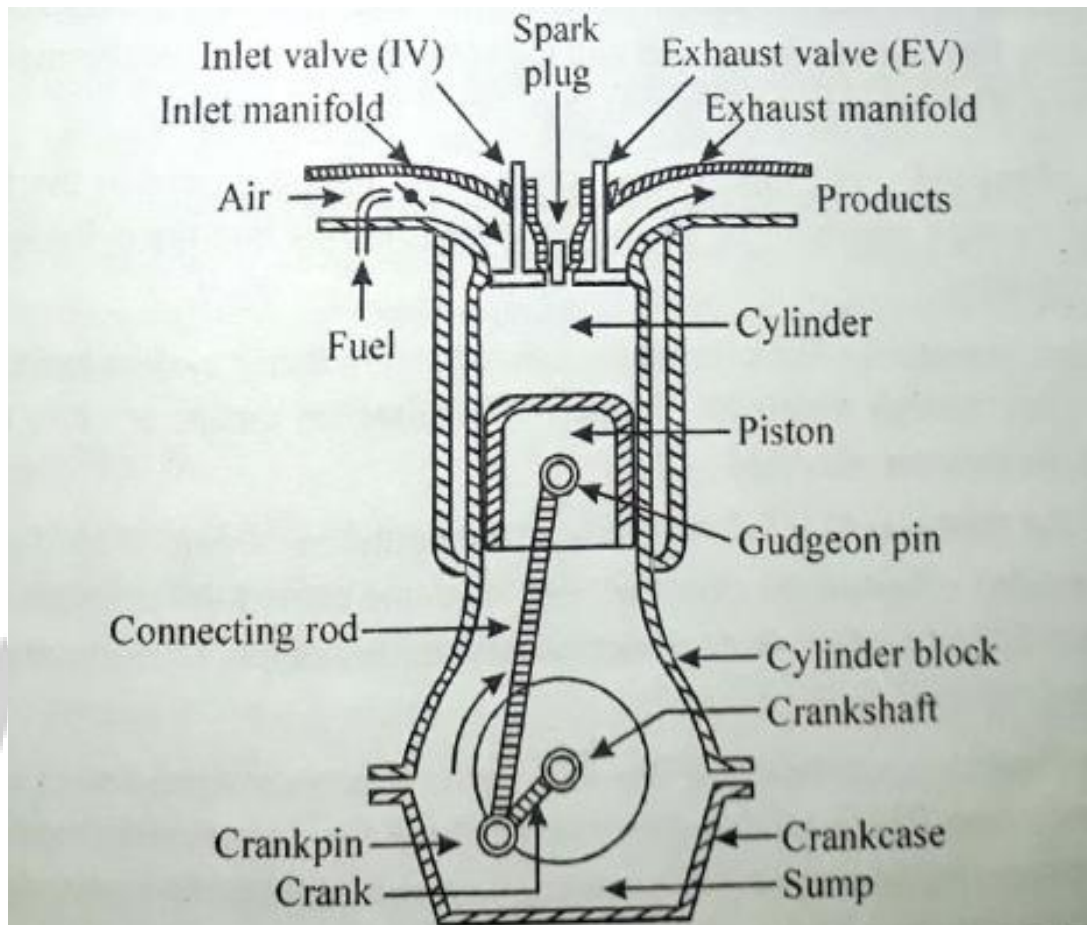


Fig.1 Cross section of SI Engine

1. Cylinder Block : It is the main supporting structure for the various components. The cylinder of a multi-cylinder engine is cast as a single unit, called cylinder block. The cylinder head is mounted on the cylinder block. The cylinder head and cylinder blocks are provided with water jackets in the case of water cooling or with cooling fins in the case of air cooling. The cylinder head is kept tight to the cylinder block by number of bolts or studs.

The bottom portion of the cylinder block is known as crankcase. The inner surface of the cylinder block which is machined and finished accurately to the cylindrical shape is called bore or face. The cylinder head is usually made of cast iron or aluminum.

2. Cylinder: It is a cylindrical vessel or space in which piston makes a reciprocating motion. The varying volume created in the cylinder during the operation of the engine is filled with the working fluid and subjected to different thermodynamic processes. The cylinder is supported in the cylinder block. It is made up of hard grade cast iron and usually cast in one piece.

3. Piston : It is a cylindrical component fitted into the cylinder forming the moving boundary of the combustion system. It fits perfectly into the cylinder providing a gas-tight space with piston rings and the lubricant. It forms the first link in transmitting the gas forces to the output shaft. Pistons are made of cast iron or aluminum alloys for lightness. Light alloy pistons expand more than cast iron therefore, they need large clearances to the bore or special provisions for expansion.

4. Combustion Chamber : The space enclosed in the upper part of the cylinder, between the cylinder head and the piston top during the combustion process, is called combustion chamber. The combustion of fuel and the consequent release of thermal energy results in increase of pressure in combustion chamber of the cylinder.

5. Inlet Manifold : The pipe which connects the intake system to the inlet valve of the engine through which air or air-fuel mixture is drawn into the cylinder is known as the inlet manifold.

6. Exhaust Manifold : The pipe which connects the exhaust system to the exhaust valve of the engine through which the products of combustion escape out into the atmosphere is called the exhaust manifold.

7. Inlet and Exhaust Valves : Valves are commonly mushroom shaped poppet type. They are provided either on the cylinder head or on the side of the cylinder for regulating the charge coming into the cylinder (inlet valve) and for discharging the products of combustion (exhaust valve) from the cylinder.

8. Spark Plug : It is a component that initiate the combustion process in Spark Ignition (SI) engines and usually located on the cylinder head. The main function of the spark plug is to provide a gap in the combustion chamber for the discharge

of a high voltage electric pulse that will ignite the air-fuel mixture at desired point in the cycle.

9. Connecting Rod: It transmits the piston load to the crank, causing the latter to turn, thus converting the reciprocating motion of the piston into a rotary motion of the crankshaft. The two ends of the connecting rod are known as small end and the big end. The small end is connected to the piston by gudgeon pin and the big end is connected to the crankshaft by crankpins. The connecting rods are made up of nickel, chromium and chromium-vanadium steels. For small engines the material may be of aluminum.

10. Crank Shaft : It converts the reciprocating motion of the piston into useful rotary motion of the output shaft. It is usually a steel forging, but some makers use special types of cast iron such as spheroidal graphitic or nickel alloy castings which are cheaper to produce and have good service life. It is enclosed in a crankcase.

11. Piston Rings : To provide a good sealing fit between the piston and the cylinder, pistons are equipped with piston rings. The rings are usually made of cast iron of fine grade and high elasticity which is not affected by the working heat. The rings can expand and contract as required.

12. Gudgeon Pin (Or Wrist Pin Or Piston Pin) : It forms a link between the small end of the connecting rod and the piston. These pins are press fit in the piston bosses of light alloy pistons when cooled. For removal or fitting, the piston should be dipped in hot water or hot oil, this expands the bosses and the pins can be removed or fitted freely without damage. It is made hollow for lightness.

13. Camshaft : The camshaft and its associated parts control the opening and closing of the two valves. This shaft also provides the drive to the ignition system. The camshaft is driven by the crankshaft through timing gears.

14. Cams : These are made as integral parts of the camshaft and are designed in such a way to open the valves at the correct timing to keep them open for the necessary duration.

15. Flywheel : In order to achieve a uniform torque, an inertia mass in the form of a wheel is attached to the output shaft. This wheel is known as flywheel. Its

function is to store available mechanical energy when it is in excess of the load requirement and to part with the same when the available energy is less than that required by the load.

3. NOMENCLATURE

(a) Cylinder Bore (d): The nominal inner diameter of the working cylinder is called the bore and is usually expressed in millimeter (mm).

(b) Piston Area (A): The area of a circle of diameter equal to the cylinder bore is known as piston area and is usually expressed in square centimeter cm^2 .

(c) Stroke (L): The linear distance along the cylinder axis between two limiting positions is called stroke. It may also be defined as the nominal distance through which a working piston moves between two successive reversals of its directions of motion. It is usually expressed in millimeter (mm).

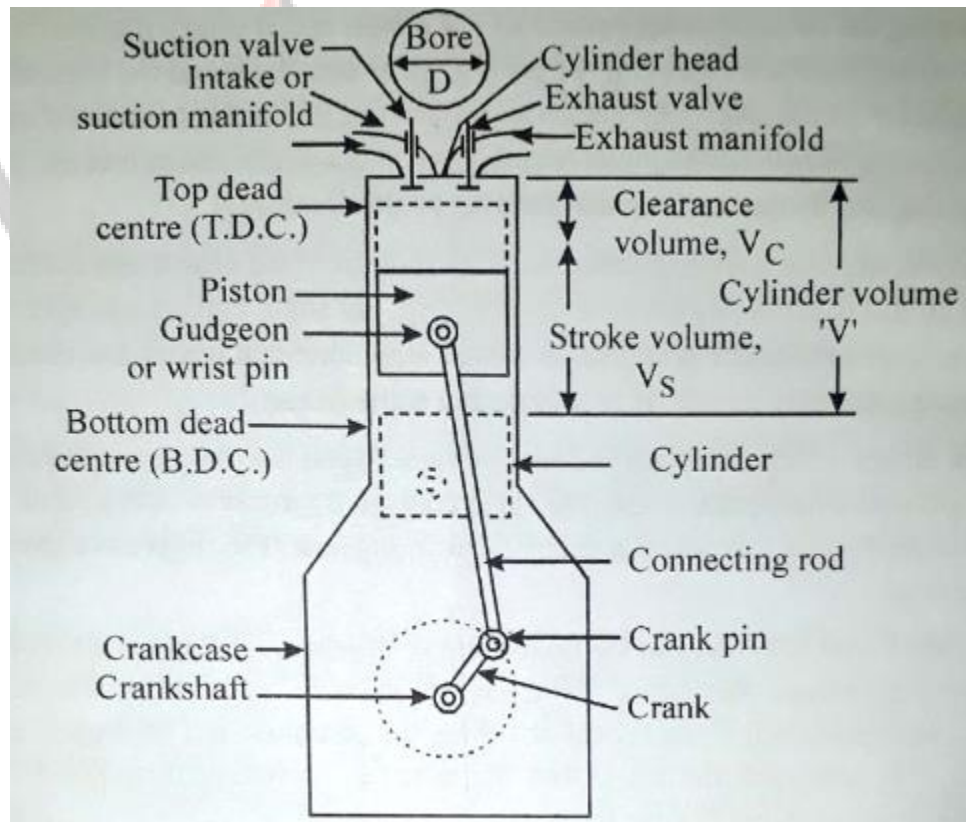


Fig.2 Different parts of SI Engine

(d) Stroke to Bore Ratio: It is an important parameter in classifying size of engine.

If $d < L$ it is called under square engine.

If $d = L$, it is called square engine.

If $d > L$, it is called over square engine.

(e) Dead Centre: The position of the working piston and the moving parts which are mechanically connected to it, at the moment when the direction of the piston motion is reversed at the either end of the stroke is called the dead center. There are two dead centers in the engine (i) Top dead centre (ii) Bottom dead centre

(i) Top Dead Centre (T.D.C.): The top most position of the piston towards cover end side of the cylinder is called top dead centre. In case of horizontal engines, this is known as inner dead centre.

(ii) Bottom Dead Centre (B.D.C.): The lowest position of the piston towards the crank end side of the cylinder is called bottom dead centre. In case of horizontal engines, this is known as outer dead centre.

(f) Displacement or Swept Volume (V_s): The volume swept through the piston, moving between TDC and BDC is known as "swept volume or piston displacement". It is expressed in terms of cubic centimeter cm^3 and is given by:

$$V_s = A \times L = \frac{\pi}{4} d^2 L$$

(g) Cubic Capacity or Engine Capacity : The displacement volume of a cylinder multiplied by number of cylinders in an engine will give the cubic capacity or the engine capacity. For example, if there are K cylinders in an engine, then:

$$\text{Cubic capacity} = V_s \times K$$

(h) Clearance Volume (V_c): -The volume contained in the cylinder above the top of the piston, when the piston is at top dead centre is known as clearance volume usually expressed in cubic centimeter (cc).

(i) Compression Ratio (r): It is the ratio of the total cylinder volume when the piston is at the bottom dead centre V_T . To the clearance volume, V_c . It is represented by letter r .

$$r = \frac{V_T}{V_c} = \frac{V_c + V_s}{V_c}$$

where V_s = Swept volume,

V_c = Clearance volume

LECTURE -22**4. CLASSIFICATION OF I.C. ENGINE: -**

Internal combustion engines may be classified as given below:

(1) According to cycle of operation

- (a) Two stroke cycle engines
- (b) Four stroke cycle engines

(2) According to cycle of combustion

- (a) Otto cycle engine (combustion at constant volume)
- (b) Diesel cycle engine (combustion at constant pressure)

(3) According to the fuel employed

- (a) Gas engine
- (b) Petrol engine
- (c) Oil engine
- (d) Kerosene engine
- (e) LPG engine
- (f) Alcohol-ethyl, methyl engine
- (g) Dual fuel and multi-fuel engines

(4) According to method of fuel supply and mixture preparation

- (a) Carbureted types
- (b) Injection types

(5) According to number of strokes

- (a) Four stroke engines
- (b) Two stroke engines

(6) According to method of ignition

- (a) Spark ignition engine
- (b) Compression ignition engine

(7) According to method of cooling in the cylinder

- (a) Air cooled engine
- (b) Water cooled engine

(8) According to cylinder arrangement

- (a) Inclined or straight engines
- (b) V-engines
- (c) Radial engines
- (d) Opposed engines

(9) According to valve or port design and location

- (a) Overhead engines
- (b) Side valve engines

(10) According to number of cylinders

- (a) single cylinder engine
- (b) Multi-cylinder engine

(11) According to method of governing

- (a) Hit and miss governed engine
- (b) Quality governed engine
- (c) Quantity governed engine

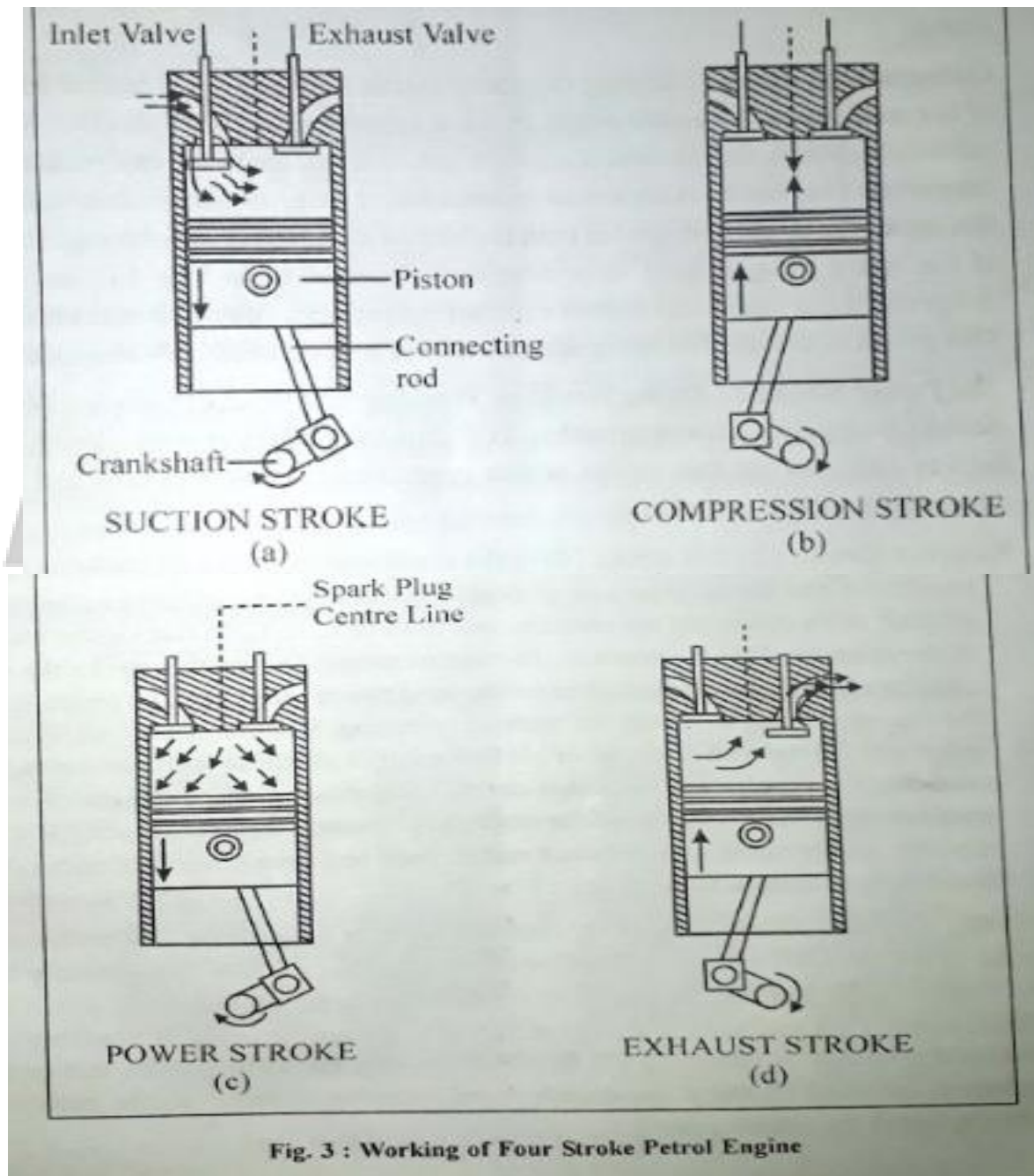
(12) According to their application

- | | |
|------------------------|--|
| (a) Stationary engines | (b) Automotive engines |
| (c) Marine engines | (d) Aircraft engines |
| (e) Industrial engines | (f) Prime movers for electrical generators |

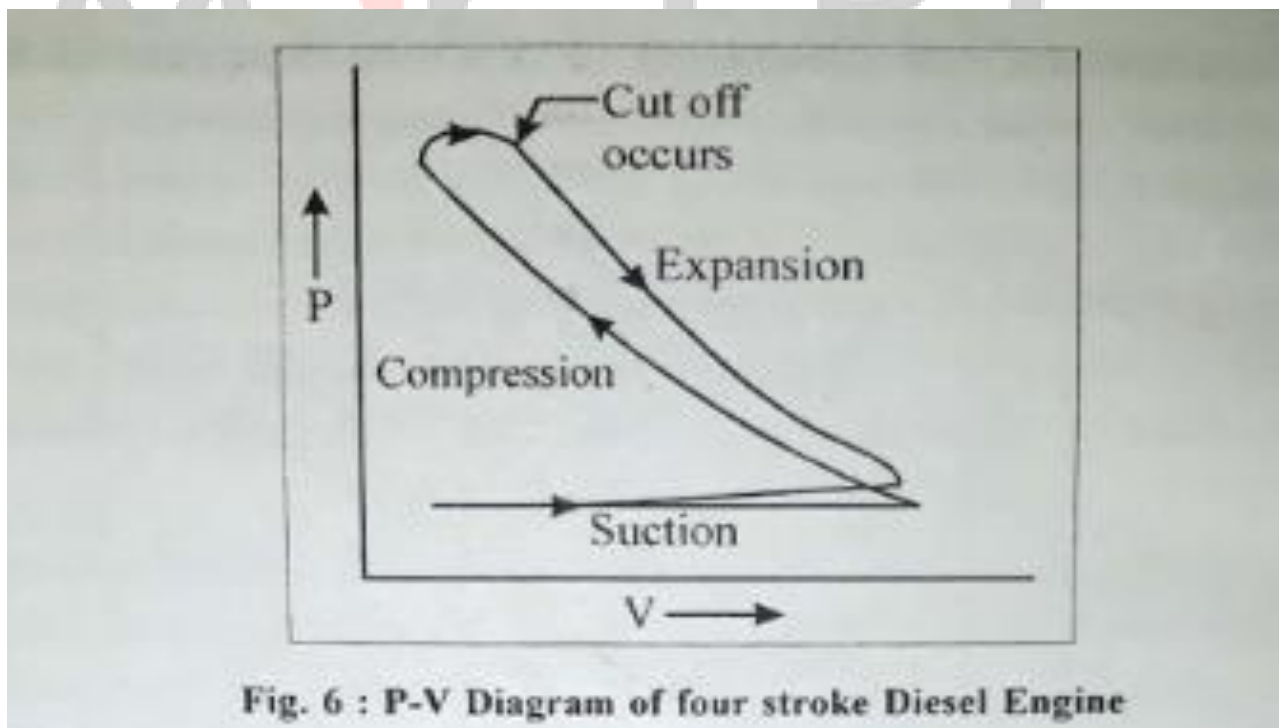
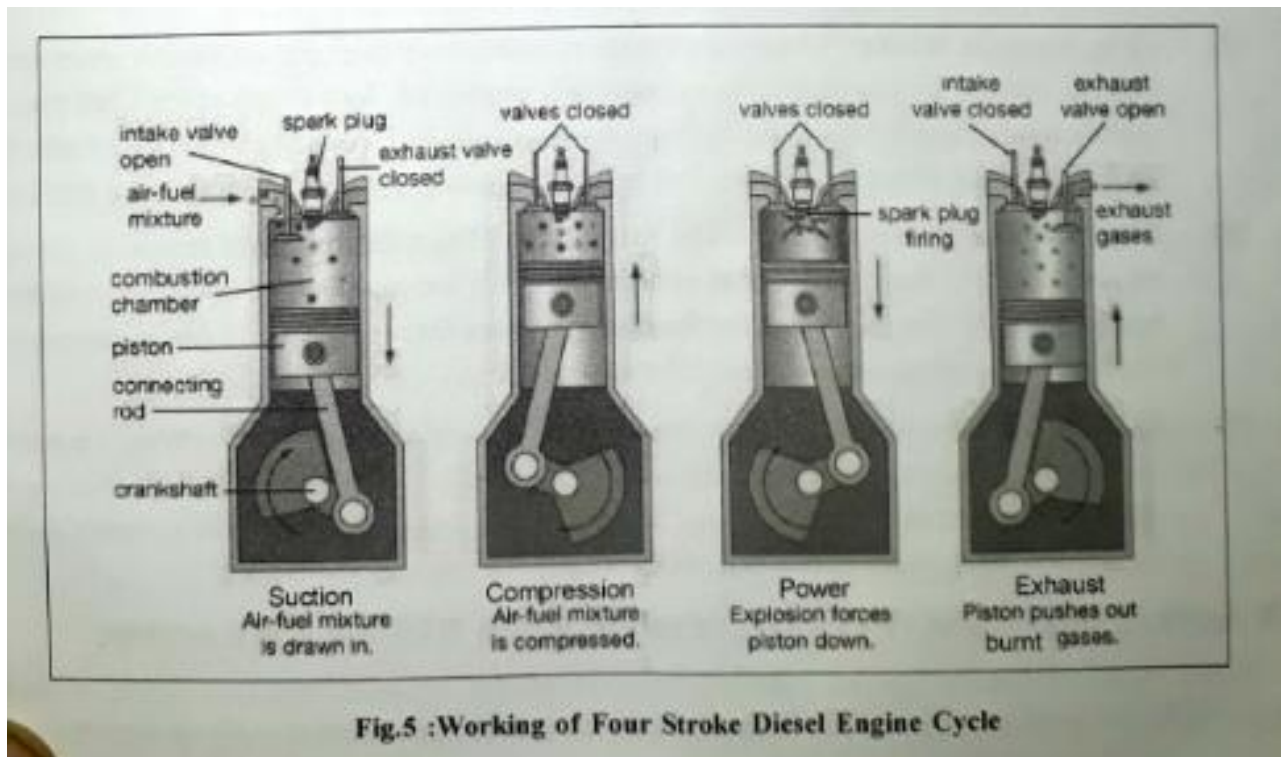
5. FOUR STROKES I.C. ENGINE

5.1 FOUR STROKE SPARK-IGNITION (S.I.) ENGINE CYCLE

In four stroke cycle engines, the cycle of operation is completed in four stroke of the piston or two revolutions of the crankshaft as discussed earlier. The four-stroke cycle principle can be described as follows and p-V diagram is also shown.



1. **Suction Stroke (Intake Stroke):** During the suction stroke (0-1), the piston travels downward in the cylinder from TDC to BDC, creating a partial vacuum in the cylinder. Atmospheric pressure forces air through the carburetor where it mixes with petrol and enters the cylinder through the intake manifold and the open intake valve. The exhaust valve remains closed during this stroke and the crankshaft turns through half a revolution. The camshaft turns through one fourth of a revolution. The engine operating temperature is somewhere between 82° and 116° . At the end of the suction stroke, inlet valve closes.
2. **Compression Stroke:** During the compression stroke (1-2), a further half revolution of the crankshaft causes the piston to move upwards from BDC to TDC. Since both the valves are closed, the mixture is now compressed into the small cavity remaining above the piston. This cavity is known as combustion or firing chamber. Just before the end of this stroke the mixture is ignited with the help of an electric spark between the electrodes of the spark plug located in combustion chamber wall. The friction between the compressed gas molecules causes a further temperature rise of several hundred degrees. This prepares the gas for firing and pressure is also considerably increased.
3. **The Power or Firing or Working or Expansion Stroke:** In this stroke (3-4), piston now approaches TDC and both valves remain closed. Of the four strokes only during this stroke power is produced. Both pressure and temperature decrease during expansion.
4. **Exhaust Stroke:** In this stroke (4-0), the crankshaft has now rotated one- and one-half revolution and the cylinder is now filled with burnt gases which must be removed. The exhaust valve opens and the piston moves from BDC to TDC, forcing the burnt gases out of the cylinder. This is known as the exhaust stroke. During this stroke the intake valve remains closed. The crankshaft has completed two revolutions. The piston is at TDC and the engine is ready to repeat the cycle of operation. Simultaneously emptying of burned gases and filling with fresh air or air-fuel mixture is known as "scavenging." Since the crankshaft makes half a revolution during each stroke, the four strokes of one complete cycle equal to two revolution of the crankshaft. Because each valve opens only once during the four-stroke cycle, the camshaft makes only one revolution. Therefore, the ratio of crankshaft to camshaft speed is 2: 1.

5.2 FOUR STROKE COMPRESSION-IGNITION ENGINE CYCLE: -

The four-stroke engine is similar to the four stroke SI engine except that high compression ratio is used in CI engines and during the suction stroke, air alone instead of a fuel-air mixture is inducted. The compression ratio of an SI engine is between 6 and 10 while for CI engine it is from 16 to 20. Due to this high compression ratio, the temperature at the end of the compression stroke is sufficient to ignite the fuel which is injected into the combustion chamber.

The actual construction of CI engine is similar to SI engine except that CI engine does not have a carburetor or spark plug. The spark plugs are replaced by an injector through which a pump forces a very fine spray of fuel.

- (1) **Suction Stroke:** The inlet valve opens and the piston moves downward, a depression is created in the cylinder. Atmospheric pressure outside the cylinder forces air through the open inlet valve into the cylinder. Once the piston has reached the end of this stroke the inlet valve is closed.
- (2) **Compression Stroke:** The piston moves upward, both inlet and exhaust valves remain closed, and the air trapped in the cylinder is compressed. As a result of the high pressure the air temperature within the cylinder rises up to 800°C . A major quantity of atomized fuel is injected into the cylinder just before the piston reaches the top of its stroke.
- (3) **Expansion or Power Stroke:** The fuel injected has a self-ignition temperature in the region of 400°C therefore, being introduced into a temperature of 800°C , it ignites and begins to burn. The expansion of the burning gases forces the piston down on its power stroke. Both valves remain closed during expansion stroke.
- (4) **Exhaust Stroke:** As the piston comes near the end of its downward stroke, the exhaust valve opens. The exhaust gases are forced out of the cylinder as the piston moves upward on its exhaust stroke. When the piston reaches the top of its travel, the exhaust valve is closed and inlet valve opens and another cycle of operations begins.

LECTURE -23**6. TWO STROKE CYCLE ENGINES**

In two stroke cycle engines, the whole sequence of events i.e., suction, compression, power and exhaust are completed in two strokes of the piston i.e., one revolution of the crankshaft. There is no valve in this type of engine. Gas movement takes place through holes called ports in the cylinder. The crankcase of the engine is air tight in which the crankshaft rotates.

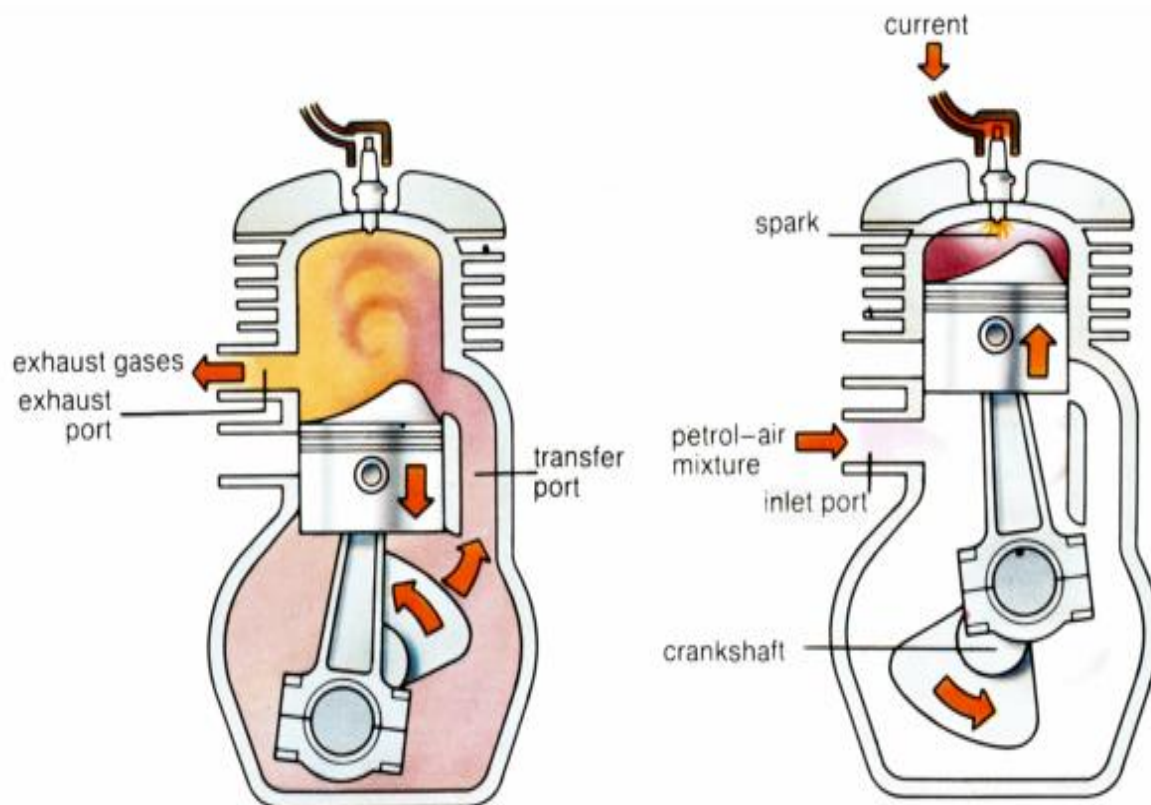
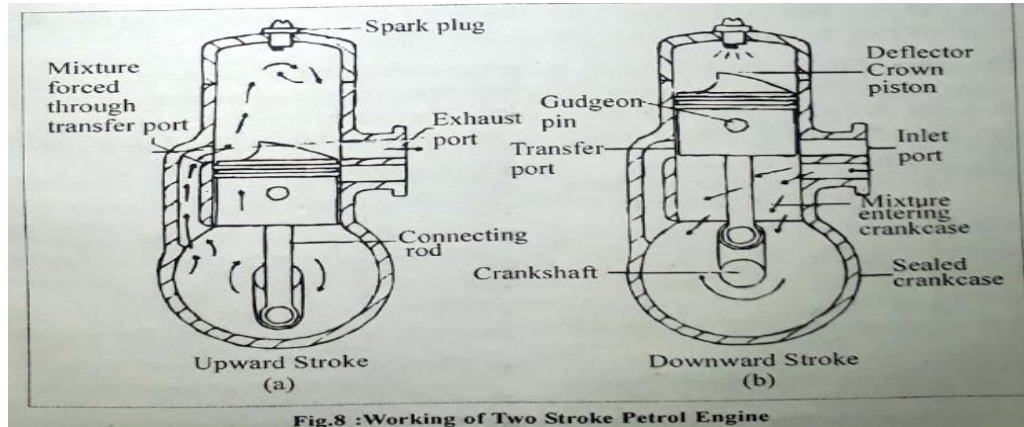


Fig. 3. Cycle of operation in two stroke engine

6.1 Working of 2-Stroke S.I. (Petrol) ENGINE

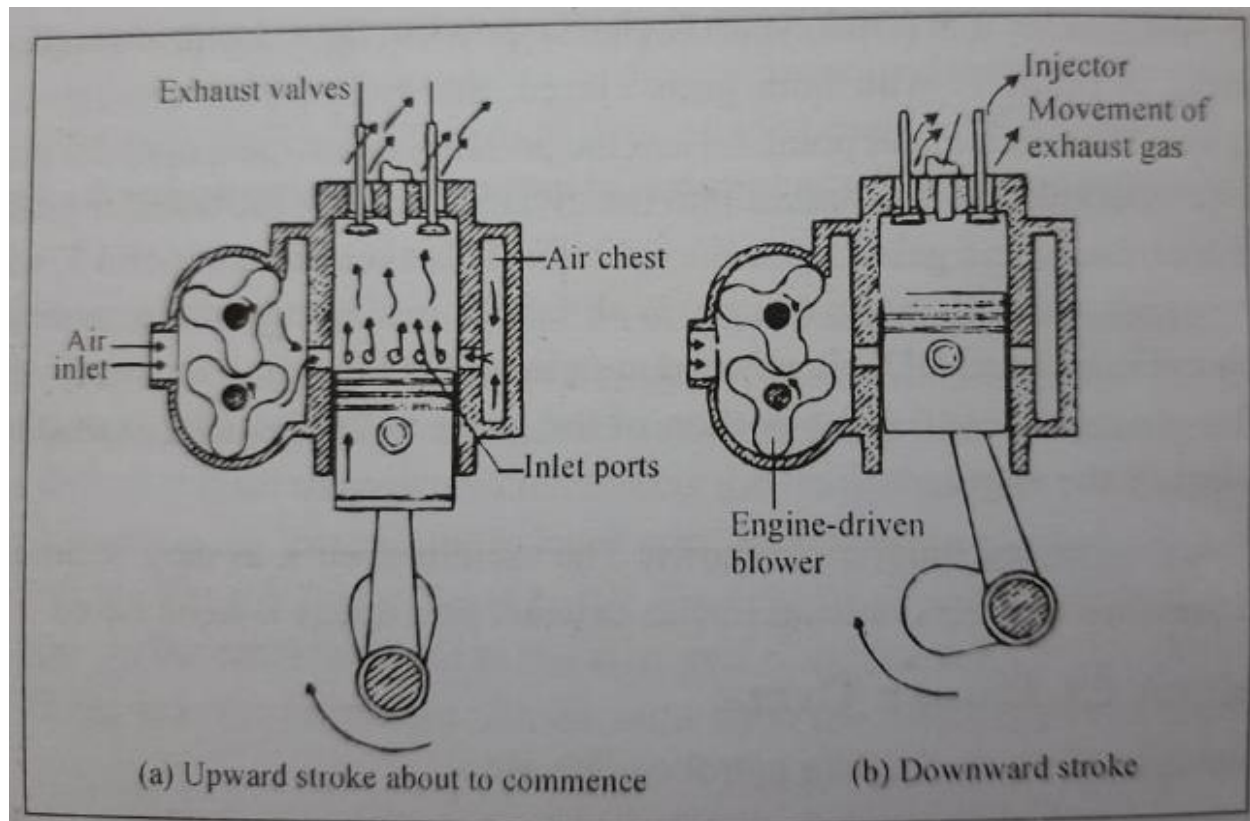


Upward stroke of the piston (Suction + Compression)

When the piston moves upward it covers two of the ports, the exhaust port and transfer port, which are normally almost opposite to each other. This traps the charge of air- fuel mixture drawn already in to the cylinder. Further upward movement of the piston compresses the charge and also uncovers the suction port. Now fresh mixture is drawn through this port into the crankcase. Just before the end of this stroke, the mixture in the cylinder is ignited by a spark plug. Thus, during this stroke both suction and compression events are completed.

Downward stroke (Power + Exhaust)

Burning of the fuel rises the temperature and pressure of the gases which forces the piston to move down the cylinder. When the piston moves down, it closes the suction port, trapping the fresh charge drawn into the crankcase during the previous upward stroke. Further downward movement of the piston uncovers first the exhaust port and then the transfer port. Now fresh charge in the crankcase moves in to the cylinder through the transfer port driving out the burnt gases through the exhaust port. Special shaped piston crown deflects the incoming mixture up around the cylinder so that it can help in driving out the exhaust gases. During the downward stroke of the piston power and exhaust events are completed.

6.2 Working of 2-Stroke C.I. (Diesel) ENGINE

Upward Stroke: Only air is allowed to pass into the cylinder while the inlet ports are uncovered, at the same time the exhaust valves are open. This combination of uncovered inlet ports and open exhaust valves permits air to be blown through the cylinder with a fresh charge of air. As the piston advances up the cylinder the inlet ports are covered, the exhaust valves are closed and the air is compressed into the lesser volume of the combustion chamber where the air temperature rises to approximately 540°C and self-ignition of fuel occurs. Just before the piston reaches the top of its stroke, fuel is injected into the cylinder and the combustion started. The heat developed from the compression of air in the cylinder ignites the fuel and the expanding gases force the piston downward on its effective power stroke.

Downward Stroke: Before the piston reaches the end of its downward or power stroke, the exhaust valve opens, allowing the burnt exhaust gases to escape. Shortly afterwards the piston uncovers the air inlet ports, allowing the pressurized air in the air chest to remove the remaining exhaust gases and charge the cylinder enabling the whole cycle to start all over again. The thermal efficiency of the CI engine is approximately less. The CI engines always take in a full charge of fresh air and therefore the fuel is more completely oxidized or burnt. The engine's objectionable odour contains practically no carbon monoxide. CI engine does not produce more smoke than an SI engine. The engine speed depends solely upon the amount of fuel injected and the load imposed on the engine. The amount of fuel injected is controlled by the throttle position and an special governor.



LECTURE - 24**7. COMPARISON BETWEEN 4-STROKE & 2-STROKE ENGINES****COMPARISON BETWEEN TWO STROKE AND FOUR STROKE ENGINES**

Four stroke engine	Two stroke engine
1. One power stroke for every two revolutions of the crankshaft.	One power stroke for each revolution of the crankshaft.
2. There are inlet and exhaust valves in the engine.	There are inlet and exhaust ports instead of valves.
3. Crankcase is not fully closed and air tight.	Crankcase is fully closed and air tight.
4. Top of the piston compresses the charge.	Both sides of the piston compress the charge.
5. Size of the flywheel is comparatively larger.	Size of the flywheel is comparatively smaller.
6. Fuel is fully consumed.	Fuel is not fully consumed.
7. Weight of engine per hp is high.	Weight of engine per hp is comparatively low.
8. Thermal efficiency is high.	Thermal efficiency is comparatively low.
9. Removal of exhaust gases easy.	Removal of exhaust gases comparatively difficult.
10. Torque produced is even.	Torque produced is less even.
11. For a given weight, engine would give only half the power of two stroke engine.	For same weight, two stroke engine gives twice the power that of four stroke engine.
12. All types of speed are possible (high and low).	Mostly high speed engines are there.
13. It can be operated in one direction only.	It can be operated in both direction (clockwise and counter clockwise).

8. COMPARISON BETWEEN S.I. & C.I. ENGINES

COMPARISON OF DIESEL ENGINE WITH PETROL ENGINE

Diesel engine	petrol engine
i) It has got no carburetor, ignition coil and spark plug.	It has got carburetor, ignition coil & spark plug.
ii) Its compression ratio varies from 14:1 to 22:1	Its compression ratio varies from 5:1 to 8:1.
iii) It uses diesel oil as fuel.	It uses petrol (gasoline) or power kerosine as fuel.
iv) Only air is sucked in cylinder in suction stroke.	Mixture of fuel and air is sucked in the cylinder in suction stroke.
v) It has got 'fuel injection pump' and injector	It has got no fuel injection pump and injector, instead it has got carburetor and ignition coil.
vi) Fuel is injected in combustion chamber where burning of fuel takes places due to heat of compression.	Air fuel mixture is compressed in the combustion chamber when it is ignited by an electric spark.
vii) Thermal efficiency varies from 32 to 38%	Thermal efficiency varies from 25 to 32%
viii) Engine weight per horse-power is high.	Engine weight per horsepower is comparatively low.
ix) Operating cost is low.	Operating cost is high.
x) Compression pressure inside the cylinder varies from 35 to 45 kg/cm ² and temperature is about 500°C.	Compression pressure varies from 6 to 10 kg/cm ² and temperature is above 260°C.